PCBC H Parametric & Sequences Test (Practice A) **Name**_____ _____/16

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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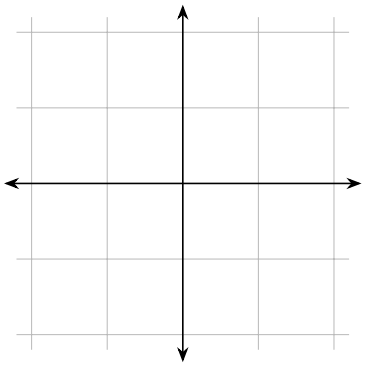
_____ _____

Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

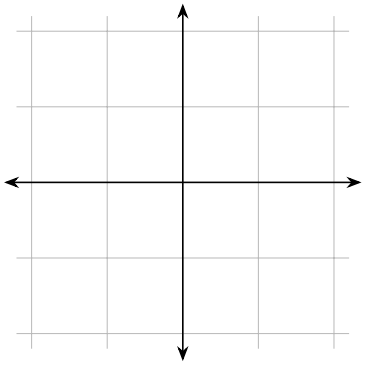
1. $x = 1 - t, \quad y = t^2 - 4t + 3, \quad -1 \le t \le 3$

SIMPLIFIED rectangular form: _____



2. $x = 2\sqrt{t}, \quad y = 8 - t, \quad 0 \le t \le 4$

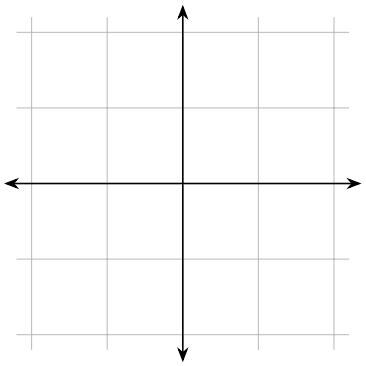
SIMPLIFIED rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

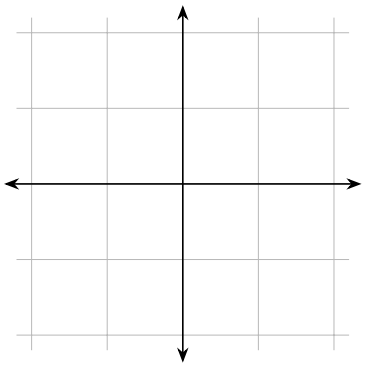
3. $x = -2\cos t, \quad y = 3\sin t, \quad 0 \le t \le \pi$

rectangular form: _____



4. $x = 3\sec t, \quad y = 2\tan t, \quad \frac{\pi}{2} < t < \frac{3\pi}{2}$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = -(x-5)^2 + 2$

A sequence is defined recursively. Write the first five terms. (2 points)

6. $a_1 = 2; \quad a_n = n\,a_{n-1} - 1$

Extra practice: the other type #6 could be

Not part of the 16 points. Worked in the key.

Write down the first five terms of the sequence.

E1. $\{a_n\} = \left\{(-1)^{n+1}\dfrac{2n}{n+3}\right\}$

Answer Key

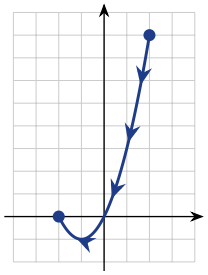
Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = 1 - t,\ y = t^2 - 4t + 3,\ -1 \le t \le 3$

t	(x, y)
-1	$(2, 8)$
0	$(1, 3)$
1	$(0, 0)$
2	$(-1, -1)$
3	$(-2, 0)$

brackets: plug in both ends



$t = 1 - x$ — *isolate t: one answer*

$y = (1 - x)^2 - 4(1 - x) + 3$ — *substitute*

$y = 1 - 2x + x^2 - 4 + 4x + 3$ — *expand; -4 times $-x$ is $+4x$*

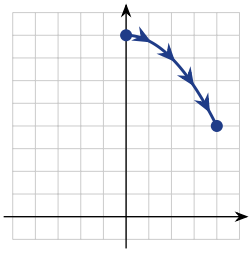
$\boxed{y = x^2 + 2x \text{ for } x \text{ in } [-2, 2]}$ — *SIMPLIFIED: y alone; domain from the chart*

starts at $(2, 8)$, moves right to left; solid dots

Check: $x = -1$ gives $1 - 2 = -1$. Matches $t = 2$.

2. $x = 2\sqrt{t},\ y = 8 - t,\ 0 \le t \le 4$

t	(x, y)
0	$(0, 8)$
1	$(2, 7)$
4	$(4, 4)$



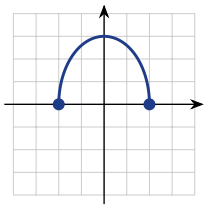
$\frac{x}{2} = \sqrt{t} \Rightarrow t = \frac{x^2}{4}$ — *divide by 2, then square*

$\boxed{y = 8 - \frac{x^2}{4} \text{ for } x \text{ in } [0, 4]}$ — *$2\sqrt{t} \ge 0$: right side only*

left to right

3. $x = -2\cos t,\ y = 3\sin t,\ 0 \le t \le \pi$

t	(x, y)
0	$(-2, 0)$
$\frac{\pi}{2}$	$(0, 3)$
π	$(2, 0)$



$\cos t = -\frac{x}{2}, \quad \sin t = \frac{y}{3}$

$\frac{x^2}{4} + \frac{y^2}{9} = 1$ — *$\sin^2 t + \cos^2 t = 1$; the minus disappears when squared*

$y^2 = 9\left(1 - \frac{x^2}{4}\right)$ — *top half only, and it uses every x from -2 to 2: a domain can't cut it, so solve for y*

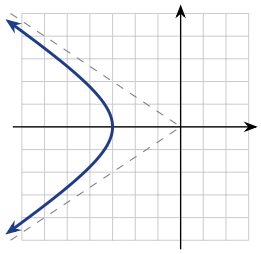
$\boxed{y = 3\sqrt{1 - \frac{x^2}{4}}}$ — *keep + (top half)*

top half only; direction not asked

4. $x = 3\sec t,\ y = 2\tan t,\ \frac{\pi}{2} < t < \frac{3\pi}{2}$

t	(x,y)
$\frac{3\pi}{4}$	$(-3\sqrt{2},-2)$
π	$(-3,0)$
$\frac{5\pi}{4}$	$(-3\sqrt{2},2)$

$\sec t < 0$ *here*



$\sec t = \frac{x}{3}, \quad \tan t = \frac{y}{2}$

$\frac{x^2}{9} - \frac{y^2}{4} = 1$ $\qquad 1 + \tan^2 t = \sec^2 t$

left branch; end arrowheads: it keeps going

$\boxed{\frac{x^2}{9} - \frac{y^2}{4} = 1 \text{ for } x \text{ in } (-\infty,-3]}$ *left branch only: a domain cuts it*

5. $y = -(x-5)^2 + 2$

$\boxed{x = t,\ y = -(t-5)^2 + 2}$

$x = t+5:\ y = -(t+5-5)^2 + 2$ *pick $t+5$ so the inside is t*

$\boxed{x = t+5,\ y = -t^2 + 2}$

Both for t in $(-\infty,\infty)$ (assumed; writing it is fine). Give both $x =$ and $y =$; don't expand.

6. $a_1 = 2;\ a_n = n\,a_{n-1} - 1$

$a_2 = 2(2) - 1 = 3$ *n times the previous term, minus 1*

$a_3 = 3(3) - 1 = 8,\ \ a_4 = 4(8) - 1 = 31$

$a_5 = 5(31) - 1 = 154$

$\boxed{2,\ 3,\ 8,\ 31,\ 154}$ *a list with commas*

Extra practice (not scored)

E1. $\left\{(-1)^{n+1}\frac{2n}{n+3}\right\}$

$\frac{2}{4},\ -\frac{4}{5},\ \frac{6}{6},\ -\frac{8}{7},\ \frac{10}{8}$ *plug in $n = 1$ to 5; odd n positive*

$\boxed{\frac{1}{2},\ -\frac{4}{5},\ 1,\ -\frac{8}{7},\ \frac{5}{4}}$ *reduce*